Arithmetic Operators (Q1–Q3)
1. Write a Java program to find the area of a triangle given its three sides using Heron’s formula.

```
import java.util.Scanner;

public class HeronsFormula {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first side: ");
        double a = sc.nextDouble();

        System.out.print("Enter second side: ");
        double b = sc.nextDouble();

        System.out.print("Enter third side: ");
        double c = sc.nextDouble();

        // Calculate semi-perimeter
        double s = (a + b + c) / 2;

        // Calculate area using Heron's formula
        double area = Math.sqrt(s * (s - a) * (s - b) * (s - c));

        System.out.println("Area of Triangle = " + area);

        sc.close();
    }
}
```

OUTPUT:
Enter first side: 3
Enter second side: 4
Enter third side: 5
Area of Triangle = 6.0

2. Create a Java program to calculate the compound interest given principal, rate, and time.

```
import java.util.Scanner;

public class CompoundInterest {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter Principal amount: ");
        double principal = sc.nextDouble();

        System.out.print("Enter Rate of interest: ");
        double rate = sc.nextDouble();

        System.out.print("Enter Time (in years): ");
        double time = sc.nextDouble();

        // Calculate amount
        double amount = principal * Math.pow((1 + rate / 100), time);

        // Calculate compound interest
        double compoundInterest = amount - principal;

        System.out.println("Compound Interest = " + compoundInterest);
        System.out.println("Total Amount = " + amount);

        sc.close();
    }
}
```

OUTPUT:
Enter Principal amount: 10000
Enter Rate of interest: 10
Enter Time (in years): 2
Compound Interest = 2100.0
Total Amount = 12100.0

3. Develop a Java program to find the distance between two points in a 2D plane using the distance formula.

```
import java.util.Scanner;

public class DistanceBetweenPoints {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter x1: ");
        double x1 = sc.nextDouble();

        System.out.print("Enter y1: ");
        double y1 = sc.nextDouble();
```

```
        System.out.print("Enter x2: ");
        double x2 = sc.nextDouble();

        System.out.print("Enter y2: ");
        double y2 = sc.nextDouble();

        double distance = Math.sqrt(
                Math.pow(x2 - x1, 2) + Math.pow(y2 - y1, 2)
        );

        System.out.println("Distance between two points = " + distance);

        sc.close();
    }
}
```

OUTPUT:
Enter x1: 2
Enter y1: 3
Enter x2: 5
Enter y2: 7
Distance between two points = 5.0

Unary Operators (Q4–Q5)
4. Write a Java program to demonstrate prefix and postfix increment by simulating a counter for visitors entering and leaving a store.

```
public class VisitorCounter {
    public static void main(String[] args) {

        int visitors = 10;

        System.out.println("Initial visitors: " + visitors);

        // Prefix increment
        System.out.println("Visitor entering using prefix increment: " + (++visitors));

        // Postfix increment
        System.out.println("Visitor count before postfix increment: " + (visitors++));
        System.out.println("Visitor count after postfix increment: " + visitors);

        // Visitor leaving
        visitors--;
        System.out.println("After one visitor leaves: " + visitors);
```

```
    }
}
```

OUTPUT:
Initial visitors: 10
Visitor entering using prefix increment: 11
Visitor count before postfix increment: 11
Visitor count after postfix increment: 12
After one visitor leaves: 11

5. Create a Java program that takes an integer and outputs its negative value using unary operators.

```
import java.util.Scanner;

public class NegativeValue {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter an integer: ");
        int number = sc.nextInt();

        int negativeValue = -number;

        System.out.println("Original number: " + number);
        System.out.println("Negative value: " + negativeValue);

        sc.close();
    }
}
```

OUTPUT:
Enter an integer: 25
Original number: 25
Negative value: -25

Assignment Operator (Q6–Q7)
6. Write a Java program to gradually reduce a number by half using /= until it becomes less than 1, counting the number of steps.

```
import java.util.Scanner;

public class ReduceByHalf {
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        double number = sc.nextDouble();

        int steps = 0;

        while (number >= 1) {
            number /= 2;
            steps++;
            System.out.println("After step " + steps + ": " + number);
        }

        System.out.println("Total steps: " + steps);

        sc.close();
    }
}
```

OUTPUT:
Enter a number: 10
After step 1: 5.0
After step 2: 2.5
After step 3: 1.25
After step 4: 0.625
Total steps: 4

7. Create a Java program to accumulate rainfall data for 7 days using += and find the total.

```
import java.util.Scanner;

public class RainfallData {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        double totalRainfall = 0;

        for (int day = 1; day <= 7; day++) {
            System.out.print("Enter rainfall for day " + day + " (in mm): ");
            double rainfall = sc.nextDouble();

            totalRainfall += rainfall;
        }
```

```
        System.out.println("Total rainfall for 7 days: " + totalRainfall + " mm");

        sc.close();
    }
}
```

OUTPUT:

```
Enter rainfall for day 1 (in mm): 10
Enter rainfall for day 2 (in mm): 5
Enter rainfall for day 3 (in mm): 12
Enter rainfall for day 4 (in mm): 8
Enter rainfall for day 5 (in mm): 7
Enter rainfall for day 6 (in mm): 4
Enter rainfall for day 7 (in mm): 6
Total rainfall for 7 days: 52.0 mm
```

Relational Operators (Q8–Q9)

8. Write a Java program to check whether three given angles form a valid triangle.

```
import java.util.Scanner;

public class ValidTriangle {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first angle: ");
        int angle1 = sc.nextInt();

        System.out.print("Enter second angle: ");
        int angle2 = sc.nextInt();

        System.out.print("Enter third angle: ");
        int angle3 = sc.nextInt();

        if (angle1 > 0 && angle2 > 0 && angle3 > 0
                && angle1 + angle2 + angle3 == 180) {
            System.out.println("The angles form a valid triangle.");
        } else {
            System.out.println("The angles do not form a valid triangle.");
        }

        sc.close();
    }
```

```
}
```

OUTPUT:
Enter first angle: 60
Enter second angle: 70
Enter third angle: 50
The angles form a valid triangle.

9. Create a Java program to compare two strings for lexicographic order without using built-in compare functions (use relational operators on characters).

```
import java.util.Scanner;

public class LexicographicComparison {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first string: ");
        String str1 = sc.nextLine();

        System.out.print("Enter second string: ");
        String str2 = sc.nextLine();

        int i = 0;
        int result = 0;

        while (i < str1.length() && i < str2.length()) {
            char ch1 = str1.charAt(i);
            char ch2 = str2.charAt(i);

            if (ch1 < ch2) {
                result = -1;
                break;
            } else if (ch1 > ch2) {
                result = 1;
                break;
            }

            i++;
        }

        if (result == 0) {
            if (str1.length() < str2.length()) {
```

```
            result = -1;
        } else if (str1.length() > str2.length()) {
            result = 1;
        }
    }

    if (result == 0) {
        System.out.println("Both strings are equal.");
    } else if (result < 0) {
        System.out.println(str1 + " comes before " + str2);
    } else {
        System.out.println(str1 + " comes after " + str2);
    }

    sc.close();
  }
}
```

OUTPUT:
Enter first string: Apple
Enter second string: Banana
Apple comes before Banana

Logical Operators (Q10–Q11)
10. Write a Java program to determine if a given year is both a leap year and within a given range.

```
import java.util.Scanner;

public class LeapYearRange {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter a year: ");
    int year = sc.nextInt();

    System.out.print("Enter starting year of range: ");
    int start = sc.nextInt();

    System.out.print("Enter ending year of range: ");
    int end = sc.nextInt();

    boolean isLeapYear;
```

```
        if (year % 400 == 0) {
            isLeapYear = true;
        } else if (year % 100 == 0) {
            isLeapYear = false;
        } else if (year % 4 == 0) {
            isLeapYear = true;
        } else {
            isLeapYear = false;
        }

        if (isLeapYear && year >= start && year <= end) {
            System.out.println(year + " is a leap year and lies within the given range.");
        } else {
            System.out.println(year + " does not satisfy both conditions.");
        }

        sc.close();
    }
}
```

OUTPUT:
Enter a year: 2024
Enter starting year of range: 2020
Enter ending year of range: 2030
2024 is a leap year and lies within the given range.

11. Create a Java program to check if a student passes a course, where they must have at least 40% in theory AND 50% in practical OR overall, 50%.

```
import java.util.Scanner;

public class StudentPass {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter theory marks (out of 100): ");
        double theory = sc.nextDouble();

        System.out.print("Enter practical marks (out of 100): ");
        double practical = sc.nextDouble();

        double overall = (theory + practical) / 2;
```

```
        if ((theory >= 40 && practical >= 50) || overall >= 50) {
            System.out.println("Student has Passed.");
        } else {
            System.out.println("Student has Failed.");
        }

        System.out.println("Overall Percentage: " + overall + "%");

        sc.close();
    }
}
```

Ternary Operator (Q12–Q13)
12. Write a Java program to find the smallest of four numbers using nested ternary operators.

```
import java.util.Scanner;

public class SmallestOfFour {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        System.out.print("Enter third number: ");
        int c = sc.nextInt();

        System.out.print("Enter fourth number: ");
        int d = sc.nextInt();

        int smallest = (a < b)
                ? ((a < c) ? ((a < d) ? a : d)
                        : ((c < d) ? c : d))
                : ((b < c) ? ((b < d) ? b : d)
                        : ((c < d) ? c : d));

        System.out.println("Smallest number is: " + smallest);

        sc.close();
    }
}
```

OUTPUT:
Enter first number: 25
Enter second number: 10
Enter third number: 30
Enter fourth number: 15
Smallest number is: 10

13. Create a Java program to check if a character is a vowel, consonant, digit, or special symbol using the ternary operator.

```
import java.util.Scanner;

public class CharacterCheck {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a character: ");
        char ch = sc.next().charAt(0);

        String result = (ch >= '0' && ch <= '9')
                ? "Digit"
                : ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z'))
                    ? ((ch == 'A' || ch == 'E' || ch == 'I' ||
                        ch == 'O' || ch == 'U' || ch == 'a' ||
                        ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u')
                        ? "Vowel"
                        : "Consonant")
                    : "Special Symbol";

        System.out.println(ch + " is a " + result);

        sc.close();
    }
}
```

OUTPUT:
Enter a character: A
A is a Vowel

Bitwise Operators (Q14–Q15)
14. Write a Java program to swap two integers without using a temporary variable (use XOR).

```
import java.util.Scanner;
```

```
public class SwapUsingXOR {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        System.out.println("Before swapping:");
        System.out.println("a = " + a);
        System.out.println("b = " + b);

        // Swapping using XOR
        a = a ^ b;
        b = a ^ b;
        a = a ^ b;

        System.out.println("After swapping:");
        System.out.println("a = " + a);
        System.out.println("b = " + b);

        sc.close();
    }
}
```

OUTPUT:
Enter first number: 10
Enter second number: 20
Before swapping:
a = 10
b = 20
After swapping:
a = 20
b = 10

15. Create a Java program to count the number of set bits (1s) in the binary representation of a number using bitwise operators.

```
import java.util.Scanner;

public class CountSetBits {
```

```
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a positive integer: ");
        int number = sc.nextInt();

        int count = 0;

        while (number > 0) {
            count += number & 1;
            number = number >> 1;
        }

        System.out.println("Number of set bits (1s): " + count);

        sc.close();
    }
}
```

OUTPUT:
Enter a positive integer: 13
Number of set bits (1s): 3

Shift Operators (Q16–Q17)
16. Write a Java program to perform fast multiplication and division of a number by powers of two using shift operators.

```
import java.util.Scanner;

public class ShiftOperations {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int number = sc.nextInt();

        System.out.print("Enter power of 2: ");
        int power = sc.nextInt();

        // Multiplication by 2^power
        int multiplication = number << power;

        // Division by 2^power
        int division = number >> power;
```

```
        System.out.println(number + " multiplied by 2^" + power +
                        " = " + multiplication);

        System.out.println(number + " divided by 2^" + power +
                        " = " + division);

        sc.close();
    }
}
```

OUTPUT:
Enter a number: 20
Enter power of 2: 2
20 multiplied by 2^2 = 80
20 divided by 2^2 = 5

17. Create a Java program to cyclically rotate the bits of an integer to the left by 2 positions.

```
import java.util.Scanner;

public class BitRotation {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter an integer: ");
        int number = sc.nextInt();

        int rotatedNumber = (number << 2) | (number >>> (32 - 2));

        System.out.println("Original number: " + number);
        System.out.println("Number after left rotation by 2 positions: "
                        + rotatedNumber);

        sc.close();
    }
}
```

OUTPUT:
Enter an integer: 5
Original number: 5
Number after left rotation by 2 positions: 20

B. Loops & Control Statements for loop (Q18–Q19)

18. Write a Java program to print the first 20 terms of the Fibonacci sequence using a for loop.

```
public class FibonacciSeries {
    public static void main(String[] args) {

        int first = 0, second = 1;

        System.out.println("First 20 terms of Fibonacci sequence:");

        for (int i = 1; i <= 20; i++) {
            System.out.print(first + " ");

            int next = first + second;
            first = second;
            second = next;
        }
    }
}
```

OUTPUT:
First 20 terms of Fibonacci sequence:
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584 4181

19. Create a Java program to print all Armstrong numbers between 1 and 1000 using a for loop.

```
public class ArmstrongNumbers {
    public static void main(String[] args) {

        System.out.println("Armstrong numbers between 1 and 1000:");

        for (int number = 1; number <= 1000; number++) {
            int originalNumber = number;
            int temp = number;
            int sum = 0;

            while (temp > 0) {
                int digit = temp % 10;
                sum = sum + (digit * digit * digit);
                temp = temp / 10;
            }

            if (sum == originalNumber) {
                System.out.println(number);
            }
```

```
        }
    }
}
```

OUTPUT:
Armstrong numbers between 1 and 1000:
1
153
370
371
407

do-while loop (Q20–Q21)
20. Write a Java program to repeatedly accept a password until the correct one is entered (use do-while).

```
import java.util.Scanner;

public class PasswordCheck {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String correctPassword = "java123";
        String password;

        do {
            System.out.print("Enter password: ");
            password = sc.nextLine();

            if (!password.equals(correctPassword)) {
                System.out.println("Incorrect password. Try again.");
            }

        } while (!password.equals(correctPassword));

        System.out.println("Correct password! Access granted.");

        sc.close();
    }
}
```

OUTPUT:
Enter password: hello

Incorrect password. Try again.

Enter password: java123
Correct password! Access granted.

21. Create a Java program to display all factors of a number using a do-while loop.

```
import java.util.Scanner;

public class FactorsOfNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a positive number: ");
        int number = sc.nextInt();

        int i = 1;

        System.out.println("Factors of " + number + " are:");

        do {
            if (number % i == 0) {
                System.out.print(i + " ");
            }
            i++;
        } while (i <= number);

        sc.close();
    }
}
```

OUTPUT:
Enter a positive number: 12
Factors of 12 are:
1 2 3 4 6 12

while loop (Q22–Q23)
22. Write a Java program to reverse the digits of an integer using a while loop.

```
import java.util.Scanner;

public class ReverseNumber {
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter an integer: ");
        int number = sc.nextInt();

        int originalNumber = number;
        int reverse = 0;

        while (number != 0) {
            int digit = number % 10;
            reverse = reverse * 10 + digit;
            number = number / 10;
        }

        System.out.println("Original number: " + originalNumber);
        System.out.println("Reversed number: " + reverse);

        sc.close();
    }
}
```

OUTPUT:
Enter an integer: 12345
Original number: 12345
Reversed number: 54321

23. Create a Java program to determine if a number is a palindrome using a while loop .

```
import java.util.Scanner;

public class PalindromeCheck {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int number = sc.nextInt();

        int originalNumber = number;
        int reverse = 0;

        while (number != 0) {
            int digit = number % 10;
            reverse = reverse * 10 + digit;
```

```
            number = number / 10;
        }

        if (originalNumber == reverse) {
            System.out.println(originalNumber + " is a Palindrome Number.");
        } else {
            System.out.println(originalNumber + " is not a Palindrome Number.");
        }

        sc.close();
    }
}
```

OUTPUT:
Enter a number: 1221
1221 is a Palindrome Number.

for-each loop (Q24–Q25)
24. Write a Java program to find the maximum and minimum values in an integer array using a for-each loop.

```
import java.util.Scanner;

public class MaxMinArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();

        int[] numbers = new int[n];

        System.out.println("Enter " + n + " elements:");

        for (int i = 0; i < n; i++) {
            numbers[i] = sc.nextInt();
        }

        int max = numbers[0];
        int min = numbers[0];

        // Using for-each loop
        for (int number : numbers) {
```

```
            if (number > max) {
                max = number;
            }

            if (number < min) {
                min = number;
            }
        }

        System.out.println("Maximum value: " + max);
        System.out.println("Minimum value: " + min);

        sc.close();
    }
}
```

OUTPUT:
Enter the number of elements: 5
Enter 5 elements:
10 25 5 40 15
Maximum value: 40
Minimum value: 5

25. Create a Java program to calculate the average marks of students stored in a 2D array using a for-each loop.

```
public class AverageMarks {
    public static void main(String[] args) {

        // Marks of students in different subjects
        int[][] marks = {
            {80, 75, 90},
            {70, 85, 80},
            {90, 88, 92}
        };

        int sum = 0;
        int count = 0;

        // Using nested for-each loops
        for (int[] student : marks) {
            for (int mark : student) {
                sum += mark;
```

```
            count++;
        }
    }

    double average = (double) sum / count;

    System.out.println("Total Marks: " + sum);
    System.out.println("Average Marks: " + average);
  }
}
```

OUTPUT:
Total Marks: 750
Average Marks: 83.33333333333333

C. switch-case, continue, break switch-case (Q26–Q27)
26. Write a Java program that takes a day number (1–7) and displays whether it's a weekday or weekend using switch-case.

```
import java.util.Scanner;

public class WeekdayWeekend {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter day number (1-7): ");
    int day = sc.nextInt();

    switch (day) {
      case 1:
      case 2:
      case 3:
      case 4:
      case 5:
        System.out.println("Weekday");
        break;

      case 6:
      case 7:
        System.out.println("Weekend");
        break;

      default:
        System.out.println("Invalid day number.");
```

```
        }

        sc.close();
    }
}
```

OUTPUT:
Enter day number (1-7): 6
Weekend

27. Create a Java program that takes a grade (A–F) and prints its qualitative meaning (e.g., A → Excellent, B → Good, etc.) using switch-case.

```
import java.util.Scanner;

public class GradeMeaning {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter grade (A-F): ");
        char grade = sc.next().charAt(0);

        switch (grade) {
            case 'A':
            case 'a':
                System.out.println("Excellent");
                break;

            case 'B':
            case 'b':
                System.out.println("Good");
                break;

            case 'C':
            case 'c':
                System.out.println("Average");
                break;

            case 'D':
            case 'd':
                System.out.println("Below Average");
                break;
```

```
            case 'E':
            case 'e':
                System.out.println("Poor");
                break;

            case 'F':
            case 'f':
                System.out.println("Fail");
                break;

            default:
                System.out.println("Invalid Grade");
        }

        sc.close();
    }
}
```

OUTPUT:
Enter grade (A-F): A
Excellent

continue statement (Q28)
28. Write a Java program to print numbers from 1 to 50, skipping all numbers that are perfect squares using continue.

```
public class SkipPerfectSquares {
    public static void main(String[] args) {

        for (int i = 1; i <= 50; i++) {
            int squareRoot = (int) Math.sqrt(i);

            if (squareRoot * squareRoot == i) {
                continue;
            }

            System.out.print(i + " ");
        }
    }
}
```

OUTPUT:
2 3 5 6 7 8 10 11 12 13 14 15 17 18 19 20 21 22 23 24

26 27 28 29 30 31 32 33 34 35 37 38 39 40 41 42 43 44 45
46 47 48 50

break statement (Q29)
29. Create a Java program to generate random integers between 1 and 100 and stop when a number divisible by both 7 and 13 is found using break.

```
import java.util.Random;

public class RandomNumberBreak {
    public static void main(String[] args) {

        Random random = new Random();

        while (true) {
            int number = random.nextInt(100) + 1;

            System.out.println("Generated number: " + number);

            if (number % 7 == 0 && number % 13 == 0) {
                System.out.println(number +
                        " is divisible by both 7 and 13.");
                break;
            }
        }
    }
}
```

OUTPUT:
Generated number: 25
Generated number: 56
Generated number: 78
Generated number: 91
91 is divisible by both 7 and 13.

Mixed Advanced (Q30)
30. Write a Java program that: Uses shift operators to check if a number is a power of 4, Uses bitwise to toggle the 3rd bit of that number, Uses a for loop to print its multiplication table but skips multiples of 6 using continue, Stops if it reaches a multiple of 48 using break.

```
import java.util.Scanner;

public class CombinedOperations {
```

```
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter a positive integer: ");
    int number = sc.nextInt();

    // Check whether number is a power of 4
    boolean isPowerOf4 = false;

    if (number > 0) {
        while (number % 4 == 0) {
            number = number >> 2; // Divide by 4 using right shift
        }

        if (number == 1) {
            isPowerOf4 = true;
        }
    }

    if (isPowerOf4) {
        System.out.println("The number is a power of 4.");
    } else {
        System.out.println("The number is not a power of 4.");
    }

    // Input again for further operations
    System.out.print("Enter a number for bit toggle and table: ");
    int num = sc.nextInt();

    // Toggle the 3rd bit
    int toggledNumber = num ^ (1 << 2);

    System.out.println("Number after toggling 3rd bit: " + toggledNumber);

    // Multiplication table
    System.out.println("Multiplication Table:");

    for (int i = 1; i <= 20; i++) {
        int result = num * i;

        // Skip multiples of 6
        if (result % 6 == 0) {
            continue;
        }
```

```
            System.out.println(num + " x " + i + " = " + result);

            // Stop at multiple of 48
            if (result % 48 == 0) {
                break;
            }
        }

        sc.close();
    }
}
```

OUTPUT:
Enter a positive integer: 16
The number is a power of 4.

Enter a number for bit toggle and table: 5
Number after toggling 3rd bit: 1

Multiplication Table:
5 x 1 = 5
5 x 2 = 10
5 x 4 = 20
5 x 5 = 25
...